

Problem

A 25,000/240-V rms distribution transformer has a primary current rating of 75 A.

(a) Find the transformer kVA rating.

(b) Calculate the secondary current.

Step-by-step solution

Step 1 of 2

(A) The KVA rating is,

$$\begin{aligned} S &= VI \\ &= 25,000 \times 75 = 1875 \text{ KVA} . \end{aligned}$$

Step 2 of 2

(C) Since,

$$S_1 = S_2 = V_2 I_2$$

And,

$$\begin{aligned} I_2 &= 1875 \times \frac{10^3}{240} \\ &= 7812 \text{ A} . \end{aligned}$$